Tarea Semanal 7

① $F(s) = \dfrac{s^4 + 4s^2 + 3}{s(s^2+2)} = \dfrac{(s^2+3)(s^2+1)}{s(s^2+2)}$

K_0

• sabemos que tiene K_0 (Método Foster)

$\lim_{s\to 0} F(s) = \dfrac{K_0}{s} \rightarrow \lim_{s\to 0} \not{s}\cdot\left[\dfrac{s^4+4s^2+3}{\not{s}(s^2+2)}\right] = K_0$

$\boxed{K_0 = \frac{3}{2}}$

$\lim_{s\to\infty} F(s) = K_\infty \cdot s \rightarrow \lim_{s\to\infty} \frac{1}{s}\left[\dfrac{s^4+4s^2+3}{s(s^2+2)}\right] = K_\infty$

$\boxed{K_\infty = 1}$

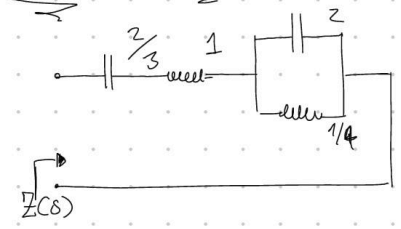
$\lim_{s^2\to(-2)} F(s) = \dfrac{2K_1 s}{s^2+2} \rightarrow \lim_{s^2\to(-2)} \dfrac{\cancel{s^2+2}}{s}\left[\dfrac{s^4+4s^2+3}{s(\cancel{s^2+2})}\right] = 2K_1$

$2K_1 = \dfrac{(-2)^2 + 4(-2) + 3}{(-2)} \Rightarrow 2K_1 = +\frac{1}{2} \rightarrow \boxed{2K_1 = \frac{1}{2}}$

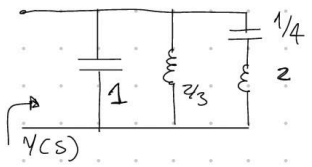
$\dfrac{2K_1 s}{s^2 + w_i^2} = \dfrac{1}{s\frac{1}{2K_1} + \frac{1}{s\cdot 2K_i/w_i^2}}$

$\boxed{\begin{array}{l} C = 2 \\ L = \frac{1}{4} \end{array}}$

$F(s) \rightsquigarrow Z(s)$



$F(s) \rightsquigarrow Y(s)$



Método cauer $F(s) = \dfrac{s^4 + 4s^2 + 3}{s(s^2+2)} = \dfrac{(s^2+3)(s^2+1)}{s(s^2+2)}$

• Remoción en infinito

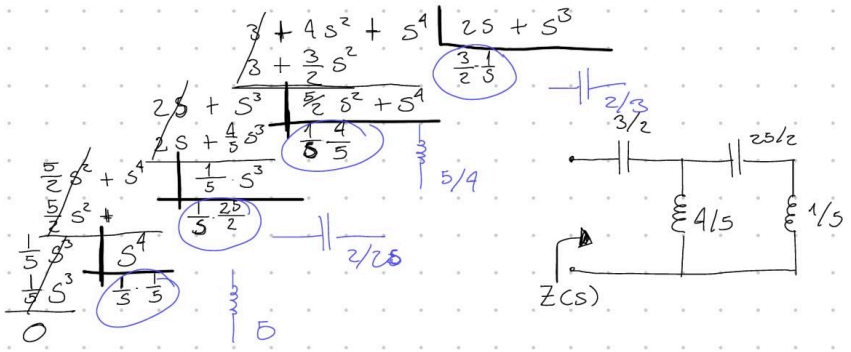
s

$F_1 = s$

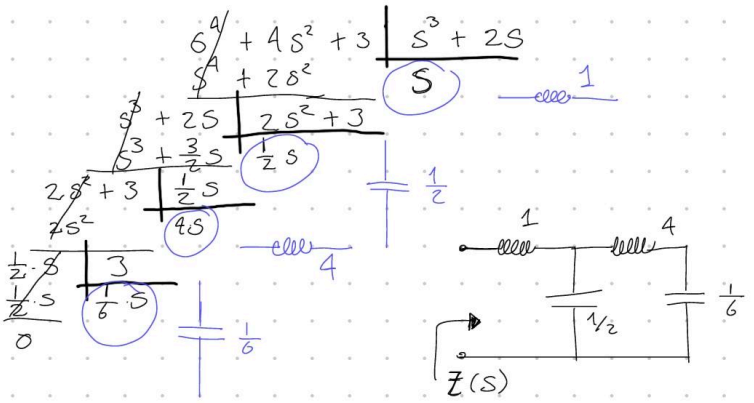
$F_2 = F - F_1 = \dfrac{s^4+4s^2+3}{s(s^2+2)} - s = \dfrac{\cancel{s^4}+4s^2+3-\cancel{s^4}+2s^2}{s}$

$F_2 = \dfrac{2s^3+4s^2+3}{s}$

Desde Huawei Notas



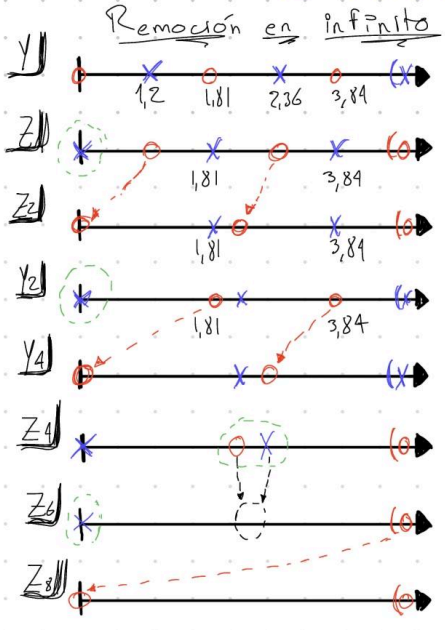
- Remoción en DC

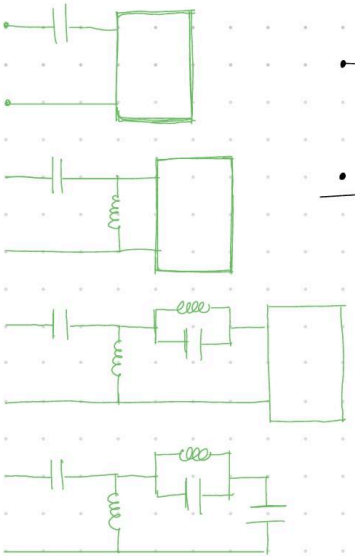


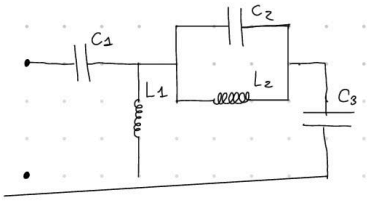
(2) $Y(s) = \frac{s^5 + 18s^3 + 48s}{6s^4 + 42s^2 + 48} = \frac{s(s^4 + 18s^2 + 48)}{6(s^4 + 7s^2 + 8)} = \frac{1}{6} \cdot \frac{s(s^2+3{,}26)(s^2+14{,}74)}{(s^2+1{,}44)(s^2+5{,}66)}$

$\frac{1}{6} \frac{s(s^2+1{,}81^2)(s^2+3{,}84)}{(s^2+1{,}2^2)(s^2+2{,}36^2)}$

Remoción en infinito







Desde Huawei Notas

$$Z = \frac{6s^4 + 42s^2 + 48}{s^5 + 18s^3 + 48s}$$

$$Z_2 = Z - \frac{K_{01}}{s} = \frac{6s^4 + 42s^2 + 48}{s^5 + 18s^3 + 48s} - \frac{1}{s}$$

$$= \frac{6s^4 + 42s^2 + 48 - s^4 - 18s^2 - 48}{s(s^4 + 18s^2 + 48)}$$

$$\boxed{Z_2 = \frac{5s^4 + 24s^2}{s(s^4 + 18s^2 + 48)} = \frac{5s^3 + 24s}{s^4 + 18s^2 + 48}}$$

$$Y_2 = \frac{s^4 + 18s^2 + 48}{5s^3 + 24s}$$

$$Y_4 = \frac{s^4 + 18s^2 + 48}{5s^3 + 24s} - \frac{K_{02}}{s}$$

$$= \frac{s^4 + 18s^2 + 48 - 10s^2 - 48}{5s^3 + 24s}$$

$$\boxed{Y_4 = \frac{s(s^3 + 8s)}{s(5s^2 + 24)} = \frac{s^3 + 8s}{5s^2 + 24}}$$

$$Z_4 = \frac{5s^2 + 24}{s(s^2 + 8)}$$

$$Z_6 = \frac{5s^2 + 24}{s^3 + 8s} - \frac{2K_{03}\,s}{s^2 + w_i^2}$$

$$= \frac{5s^2 + 24 - 2s}{s(s^2 + 8)} = \frac{3s^2 + 24}{s(s^2 + 8)}$$

$$= \frac{3(s^2 + 8)}{s(s^2 + 8)} \Rightarrow \boxed{Z_6 = \frac{3}{s}}$$

$C_3 = 1/3$

$$\lim_{s \to 0} s\,Z(s) = K_{01}$$

$$\lim_{s \to 0} \frac{6s^4 + 42s^2 + 48}{s^4 + 18s^2 + 48} = K_{01}$$

$$\boxed{K_{01} = 1 = 1/C_1}$$

$C_1 = 1$

$$\lim_{s \to 0} s\,Y_2(s) = K_{02}$$

$$\lim_{s \to 0} \frac{s^4 + 18s^2 + 48}{5s^2 + 24} = K_{02}$$

$$\boxed{K_{02} = 2 = \frac{1}{L_1}}$$

$L_1 = \frac{1}{2}$

$$\lim_{s^2 \to (-8)} \frac{s^2 + 8}{s} \cdot Y_1(s) = 2K_{03}$$

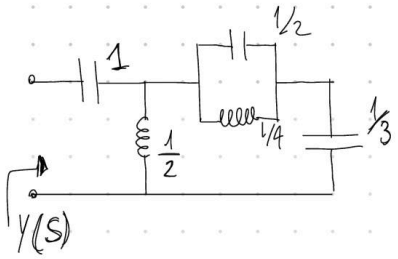
$$\lim_{s^2 \to (-8)} \frac{s^2 + 8}{s} \cdot \frac{5s^2 + 24}{s(s^2 + 8)} = 2K_{03}$$

$$\boxed{2K_{03} = 2}$$

$$\cdot \frac{2K_{03}\,s}{s^2 + w_i^2} \Rightarrow \frac{2s}{s^2 + 8} \Rightarrow \frac{1}{\frac{s^2}{2s} + \frac{8}{2s}}$$

$$\Rightarrow \frac{1}{s\left(\frac{1}{2}\right) + \frac{1}{s(1/4)}} = \frac{1}{sC + \frac{1}{sL}}$$

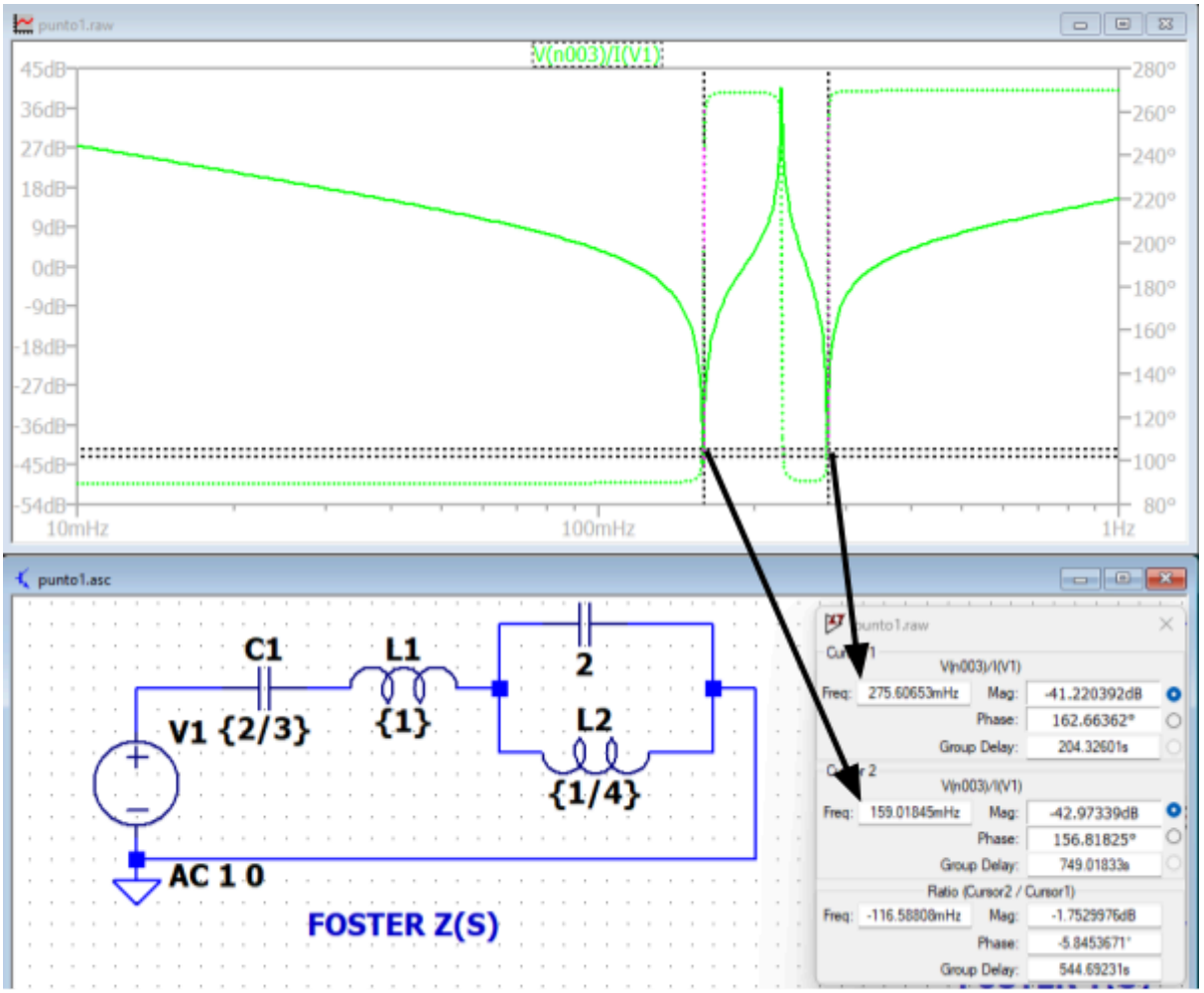
$\boxed{C = \frac{1}{2} \text{ y } L = 1/4}$



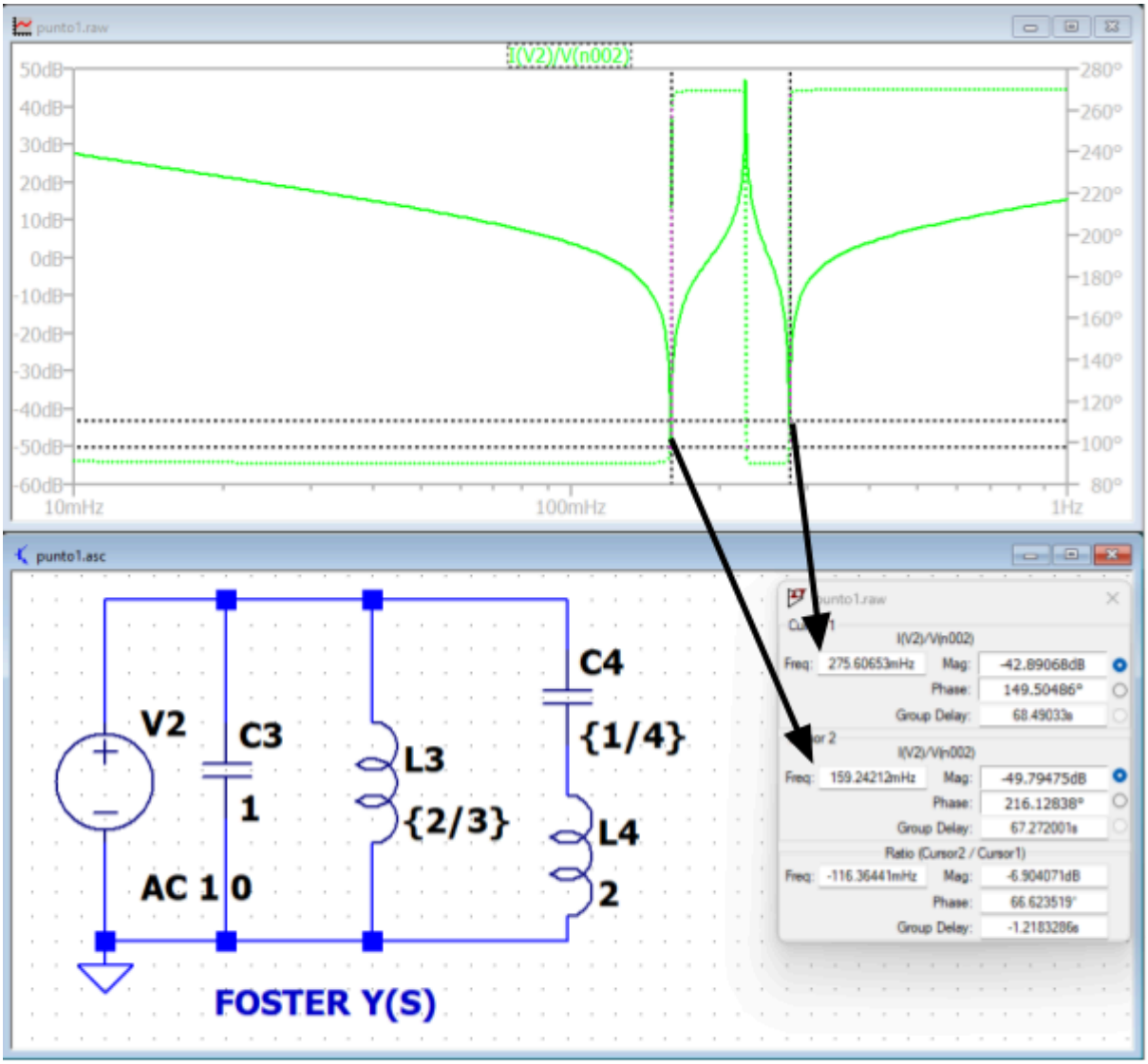
OK !!!

Desde Huawei Notas

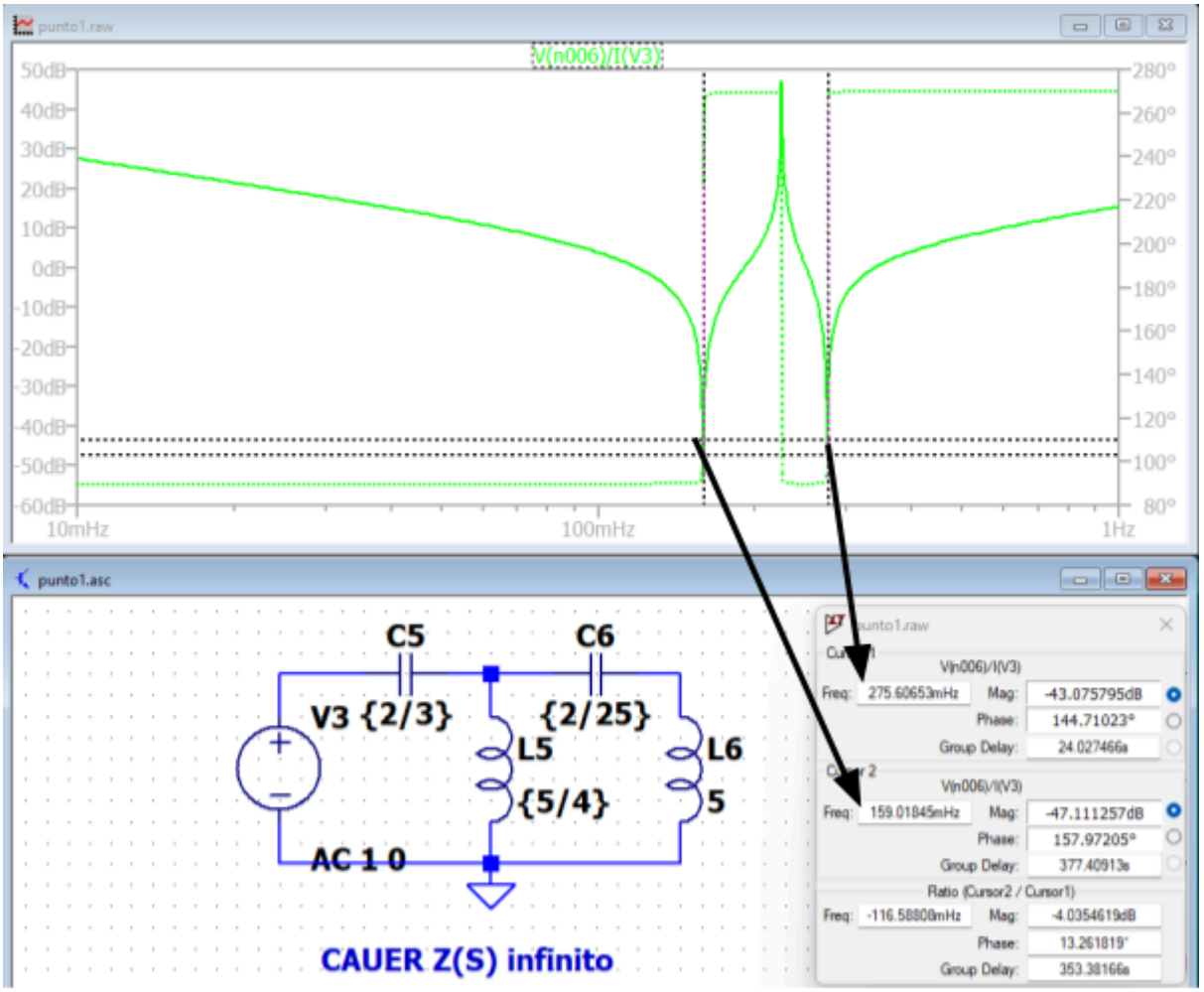
Simulación en ltSpice



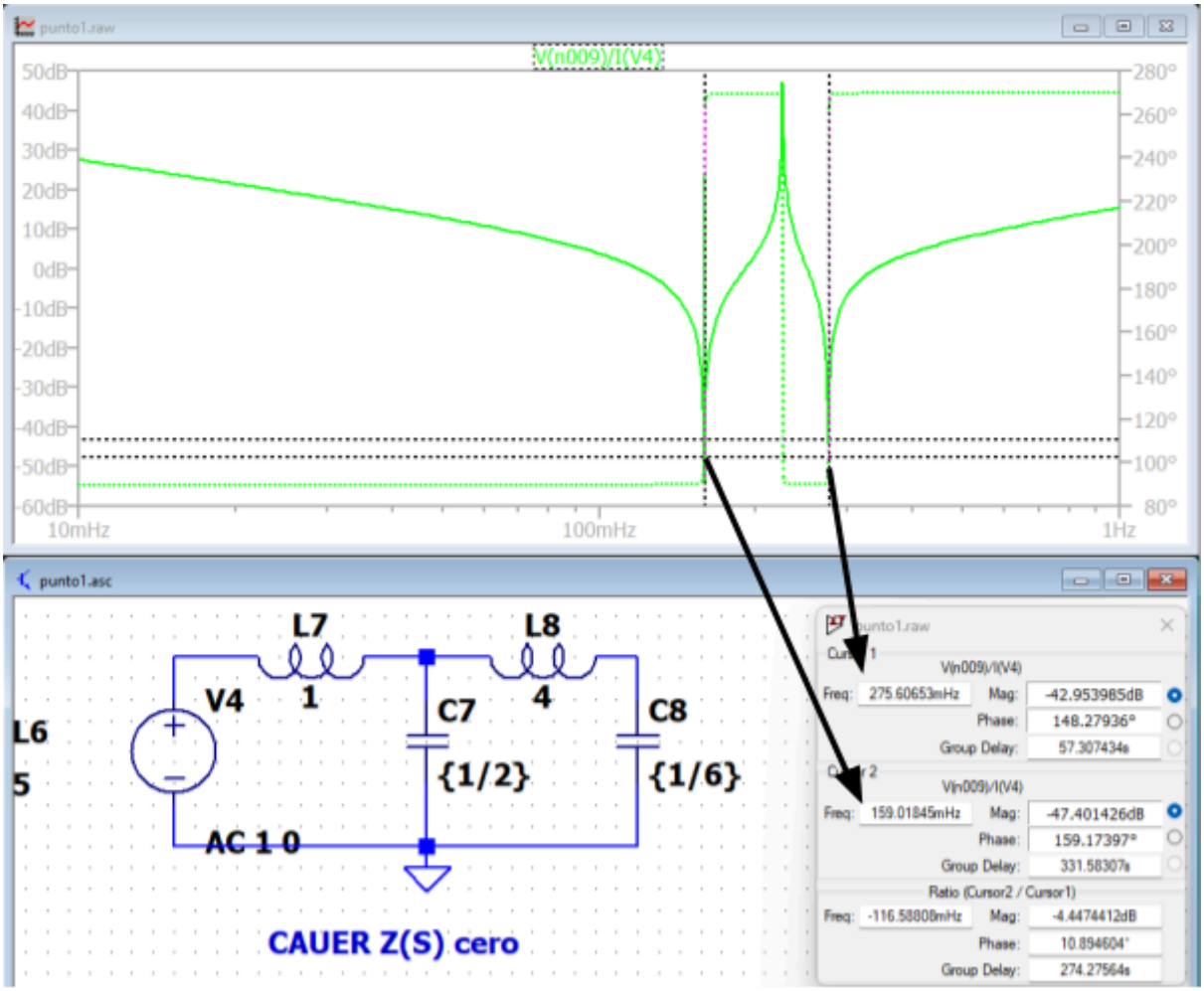
Z(s) Foster versión serie



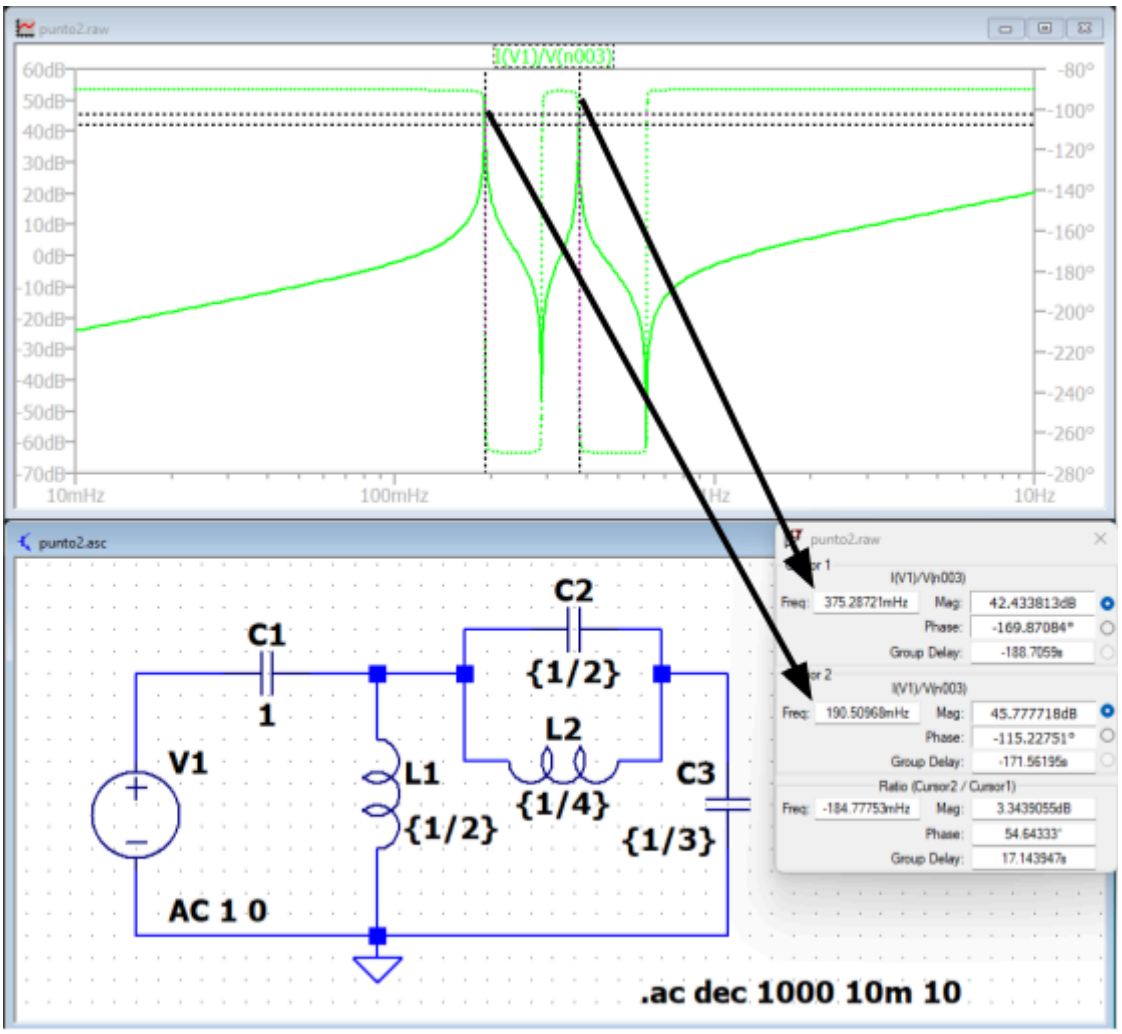
Y(s) Foster version paralelo



Z(s) Cauer remoción en infinito



Z(s) Cauer remoción en cero



Metodo grafico punto 2

Nota: respecto a las simulaciones en python, no me funciono la librería de pytc2 me falto una función o método. Adjunto error que me marca, probé el código que está de ejemplo en la página pytc2.

```
D:\documentos\proyectos_python\tc2\.venv\Scripts\python.exe
D:\documentos\proyectos_python\tc2\TS7\punto1.py
D:\documentos\proyectos_python\tc2\TS7\punto1.py:28: SyntaxWarning: invalid escape sequence '\i'
  print_latex(a_equal_b_latex_s('k_\infty', koo))
Traceback (most recent call last):
  File "D:\documentos\proyectos_python\tc2\TS7\punto1.py", line 7, in <module>
    from pytc2.general import print_latex, print_subtitle, a_equal_b_latex_s
ImportError: cannot import name 'a_equal_b_latex_s' from 'pytc2.general' (D:\documentos\proyectos_python\tc2\.venv\Lib\site-packages\pytc2\general.py)

Process finished with exit code 1
```